



Intune Cloud PKI en Entra ID Certificate-based Authentication

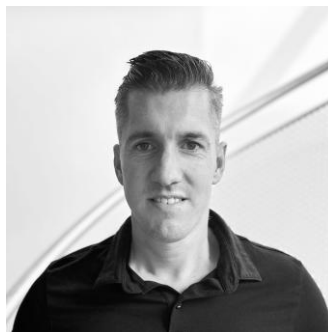
Richard van der Els | Modern Workplace Consultant

AGENDA | 25-06-2026

- 1.** Wie ben ik
- 2.** Waarom Cloud PKI
- 3.** Waarom Certificate-based Authentication
- 4.** Demo's
- 5.** Vragen?

Richard van der Els

37 jaar



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Woonplaats

's-Hertogenbosch



Samenwonend

Vriendin en dochter



Vrije tijd

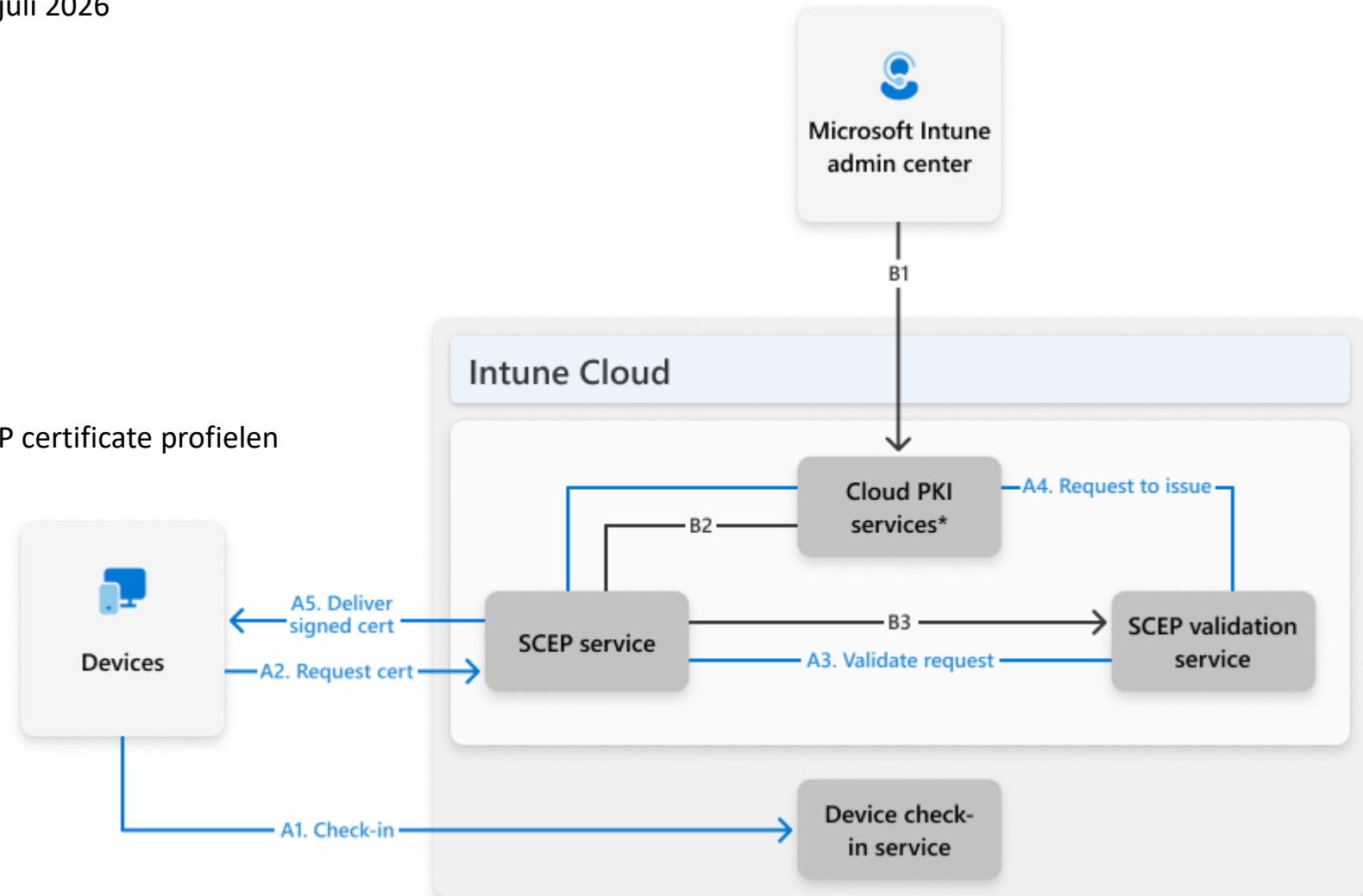
BBQ, Reizen, Uit eten

Waarom Cloud PKI

- ⌚ Onderdeel van Microsoft 365 E5 licentie vanaf juli 2026
- ⌚ Snel en eenvoudig te configureren
- ⌚ Geen complexe IT Infrastructuur benodigd
- ⌚ Naadloze integratie met Microsoft Intune

Nadelen

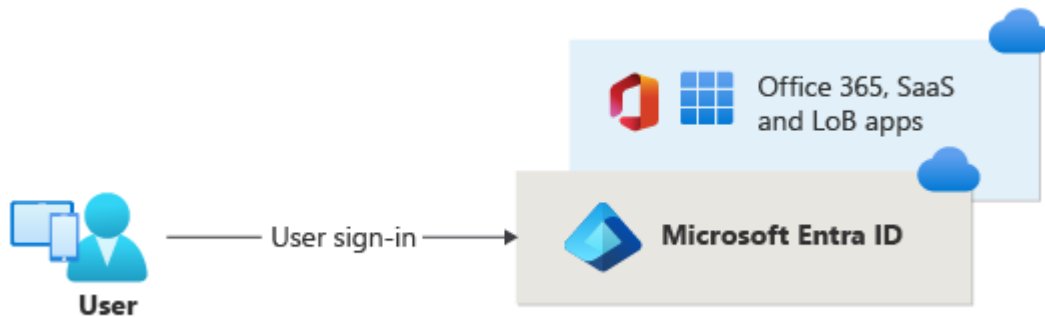
- ⌚ Alleen voor Intune Managed devices
- ⌚ Platform moet Intune device configuration SCEP certificate profielen ondersteunen
- ⌚ Max 6 CAs in één Intune tenant



Waarom Certificate-Based Authentication



- Passwordless
- Phishing resistant
- Seamless login
- Aanmelden op webtoepassingen
- Aanmelden op Office apps op iOS en Android



Bad: Password	Good: Password and...	Better: Password and...	Best: Passwordless
123456 qwerty password iloveyou Password1	 SMS Voice	 Authenticator (Push Notifications) Software Tokens OTP Hardware Tokens OTP (Preview)	 Authenticator (Phone Sign-in) Window Hello FIDO2 security key Certificates

Demo – How to setup CloudPKI

The screenshot shows the Microsoft Intune admin center interface for the 'Richard Demo Tenant'. The top navigation bar includes the 'Microsoft Intune admin center' title, a 'Copilot' button, and user information for 'richardvanderels@richar...' (RICHARD DEMO TENANT (RICHAR...)). A left-hand navigation pane lists various sections: Home, Dashboard, All services, Explorer, Devices, Apps, Endpoint security, Agents, Reports, Users, Groups, Tenant administration, and Troubleshooting + support. The main content area features a large header with the text 'Securely manage devices, access, and apps with Intune' and a sub-header 'Maximize productivity and simplify administration without compromising endpoint management and security.' Below this, a 'Status' section displays six metrics: 'Devices not in compliance' (0), 'Configuration policies with errors or conflict' (1), 'Client app install failures' (0), 'Connector errors' (0), 'Service health' (Healthy), and 'Account status' (Active). To the right, a 'Spotlight' section highlights two key features: 'Gain fleet-wide visibility into update status' (with a link to 'Explore') and 'Increase productivity with Cloud PCs' (with a link to 'Explore'). A notification banner at the top of the main content area states: 'Multi Admin Approval is recommended. Protect sensitive actions with additional approval. Learn more about Multi Admin Approval'.

Microsoft Intune admin center

Copilot

richardvanderels@richar...
RICHARD DEMO TENANT (RICHAR...)

Richard Demo Tenant

Multi Admin Approval is recommended. Protect sensitive actions with additional approval. [Learn more about Multi Admin Approval](#)

Get started

Securely manage devices, access, and apps with Intune

Maximize productivity and simplify administration without compromising endpoint management and security.

Status

Devices not in compliance 0	Connector errors 0
Configuration policies with errors or conflict 1	Service health Healthy
Client app install failures 0	Account status Active

Spotlight

Gain fleet-wide visibility into update status

The Security update status dashboard shows update readiness across Windows Client, Windows Server, and Microsoft 365 apps.

[Explore](#)

Increase productivity with Cloud PCs

Easily provision Windows 365 Cloud PCs and manage them alongside your physical devices.

[Explore](#)

Demo – Intune Trusted Certificates

The screenshot displays the Microsoft Intune admin center interface for the 'Richard Demo Tenant'. The top navigation bar includes the 'Microsoft Intune admin center' title, a 'Copilot' button, and user information for 'richardvanderels@richar...' (Richard Demo Tenant). A left-hand navigation pane lists various management areas: Home, Dashboard, All services, Explorer, Devices, Apps, Endpoint security, Agents, Reports, Users, Groups, Tenant administration, and Troubleshooting + support. The main content area features a prominent banner with the text 'Securely manage devices, access, and apps with Intune' and a sub-message: 'Maximize productivity and simplify administration without compromising endpoint management and security.' Above this banner is a notification: 'Multi Admin Approval is recommended. Protect sensitive actions with additional approval. Learn more about Multi Admin Approval'. Below the banner, there are two primary sections: 'Status' and 'Spotlight'. The 'Status' section provides a summary of system health with the following data:

Status	
Devices not in compliance	Connector errors
0	0
Configuration policies with errors or conflict	Service health
0	Healthy
Client app install failures	Account status
0	Active

The 'Spotlight' section highlights two key features:

- Gain fleet-wide visibility into update status:** The Security update status dashboard shows update readiness across Windows Client, Windows Server, and Microsoft 365 apps.
- Increase productivity with Cloud PCs:** Easily provision Windows 365 Cloud PCs and manage them alongside your physical devices.

Each spotlight item includes an 'Explore' button at the bottom.

Demo – Intune SCEP certificate

Microsoft Intune admin center

Home > Devices

Devices | Configuration

Search

Overview
All devices
Device query
Monitor
By platform
Windows
Apple mobile
macOS
Android
Linux
Device onboarding
Enrollment
Manage devices
Configuration
Compliance
Conditional access
Scripts and remediations
Group Policy analytics
eSIM cellular profiles (preview)

Policies Import ADMX Monitor

+ Create Refresh Export Columns 16 policies

Search Add filters

Policy name	Platform	Policy type	Last modified
CIS-L1-Bitlocker	Windows 10 and later	Settings catalog	3/26/2026, 2:56:18 PM
CIS-L1-Bitlocker-Startup-PIN	Windows 10 and later	Settings catalog	3/26/2026, 2:55:40 PM
CIS-L1-Defender	Windows 10 and later	Settings catalog	3/26/2026, 3:53:05 PM
CIS-L1-Intune-Base	Windows 10 and later	Settings catalog	3/26/2026, 3:53:36 PM
CIS-L1-Intune-Customizations	Windows 10 and later	Settings catalog	3/26/2026, 3:54:03 PM
CIS-L1-Microsoft-Edge	Windows 10 and later	Settings catalog	3/26/2026, 2:44:32 PM
CIS-L1-Microsoft-Office-Enterprise	Windows 10 and later	Settings catalog	3/26/2026, 2:46:45 PM
CIS-L1-Microsoft-Office-Enterprise-	Windows 10 and later	Settings catalog	3/26/2026, 2:51:45 PM
CIS-L1-Windows-Hello-for-Business	Windows 10 and later	Settings catalog	3/26/2026, 3:54:37 PM
CIS-L1-Windows-Hello-for-Business	Windows 10 and later	Settings catalog	4/21/2026, 10:35:11 AM
Microsoft-Edge-Customizations	Windows 10 and later	Settings catalog	3/26/2026, 2:46:15 PM
Microsoft-Office-Enterprise-Recomi	Windows 10 and later	Settings catalog	3/26/2026, 2:55:10 PM
Win11 - LAPS	Windows 10 and later	Local admin password solution (Win	3/26/2026, 4:03:15 PM
Win11 - Recommended Security Set	Windows 10 and later	Settings catalog	3/26/2026, 3:56:28 PM

Add or remove favorites by pressing Ctrl+Shift+F

Check issued certificates

Controleer uitgegeven certificaten in Issued leaf certificates

Home > Tenant admin | Cloud PKI | Cloud PKI > Issuing CA

Issued leaf certificates ...

Refresh Columns ▾

Search

Add filters

Subject name ↑	User principal name	Device na...	Status	Thumbprint	Serial number	Issuance	Expiration
CN=richardvanderels, E=richardvan	richardvanderels@richar...	VM-32432...	Active	F2F7B3B0176B...	0098A8D7FBCE...	15/6/2026, 12:35:	15/6/2027, 12:45:

Controleer uitgegeven certificaten op het apparaat in personal certificates

Console Root	Issued To	Issued By	Expiration Date	Intended Purposes
Certificates - Current	richardvanderels	Issuing CA	15/06/2027	Client Authentication
Personal	S-1-12-1-1964817465-13317769...	S-1-12-1-1964817465-1331776980...	05/05/2056	Smart Card Log-on
Certificates				
Trusted Root Certificates				
Enterprise Trust				
Intermediate Certificates				
Active Directory				
Trusted Publishers				
Untrusted Certificates				

Demo – Check and revoke certificate

Microsoft Intune admin center

Copilot

8

?

richardvanderels@richar...
RICHARD DEMO TENANT (RICHAR...)

Home

Dashboard

All services

Explorer

Devices

Apps

Endpoint security

Agents

Reports

Users

Groups

Tenant administration

Troubleshooting + support

Richard Demo Tenant

Multi Admin Approval is recommended. Protect sensitive actions with additional approval. [Learn more about Multi Admin Approval](#)

Get started

Securely manage devices,
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Status

Devices not in compliance
0

Configuration policies with errors or conflict
0

Client app install failures
0

Connector errors
0

Service health
Healthy

Account status
Active

Spotlight

Gain fleet-wide visibility into update status

The Security update status dashboard shows update readiness across Windows Client, Windows Server, and Microsoft 365 apps.

Explore

Increase productivity with Cloud PCs

Easily provision Windows 365 Cloud PCs and manage them alongside your physical devices.

Explore

Demo – Create Entra PKI

Microsoft Entra admin center

Search resources, services, and docs (G+)

Copilot

3

richardvanderels@richar...
RICHARD DEMO TENANT (RICHAR...

Entra ID

Overview

Users

Groups

Devices

Agents

Enterprise apps

App registrations

Roles & admins

Delegated admin partners

Tenant governance (Preview)

Domain services

Conditional Access

Multifactor authentication

Identity Secure Score

Authentication methods

Account recovery

Password reset

Custom security attributes

Microsoft Entra

Richard Demo Tenant

Tenant ID 366b70d6-9258-...
Primary domain richardin...

15
View users

15
View groups

1
View devices

4
View apps

Richard van der Els

Global Administrator

751cbc39-4dd4-4f61-a8fe-c...

View user profile

My role assignments

1

High privileged role assignments

Manage my roles

Users at high risk

No detections found

No user detections with risk level "high" in the last 365 days.

View high risk users

Tenant status

Identity Secure Score 65.40%

View recommendations

Microsoft Entra Connect

Enabled

View Entra Connect

Shortcuts

Add

User sign-ins

Audit logs

Authentication Methods

Blocked users

Domain names

Unused service principals

Manage tenants

Named locations

Cross-tenant Access Policies

Tenant restrictions

Risk Based Conditional Access policies

Risky sign-ins

Lifecycle workflows

Get the most out of your licenses and subscriptions

Deployment guides

Microsoft Entra plan

Demo – Configure Authentication Method CBA

Microsoft Entra admin center

Search resources, services, and docs (G+)

Copilot

7

richardvanderels@richar...
RICHARD DEMO TENANT (RICHAR...)

Home

Security Copilot agents

Security Copilot chat

Favorites

Entra ID

Overview

Users

Groups

Devices

Agents

Enterprise apps

App registrations

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Tenant governance (Preview)

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Multifactor authentication

Microsoft Entra

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View users

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View devices

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View apps

Richard van der Els
Global Administrator

751cbc39-4dd4-4f61-a8fe-c...

View user profile

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Risk Based Conditional Access policies

Risky sign-ins

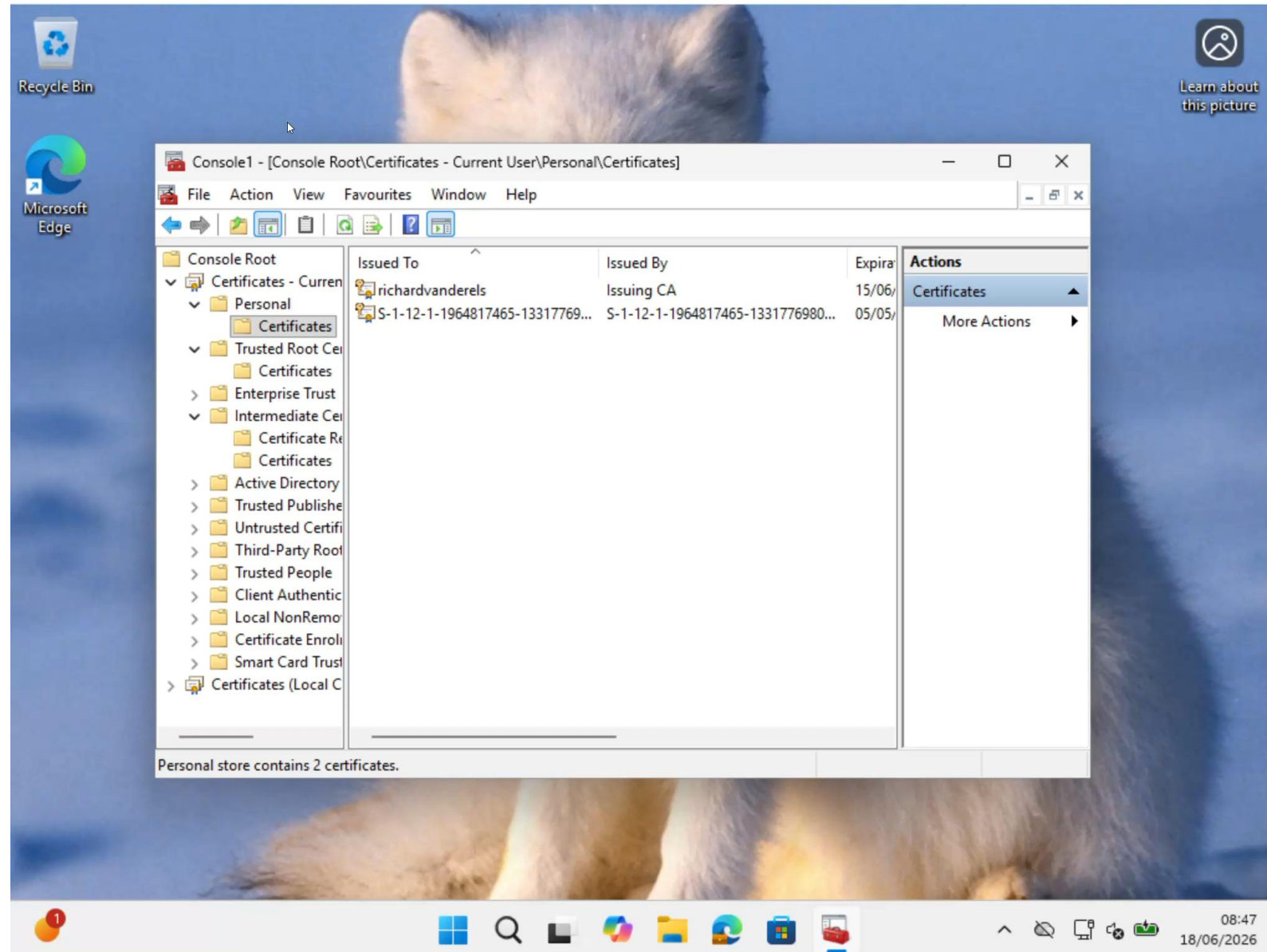
Lifecycle workflows

Get the most out of your licenses and subscriptions

Deployment guides

Microsoft Entra plan

Demo – Login met certificaat



Sign-in logs

Home > Users > Richard van der Els		
Richard van der Els Sign-in logs		
User		
Search		
Download Export Data Settings		
Overview Audit logs Sign-in logs Diagnose and solve problems Custom security attributes Assigned roles Administrative units Groups Applications Licenses Devices Azure role assignments Authentication methods New support request		
This view will soon be replaced with a view that shows all sign-in activity for the selected user.		
Date: Last 24 hours Show dates as: Last 24 hours		
User sign-ins (interactive) User sign-ins (non-interactive)		
Date	Request ID	
3/5/2026, 3:19:54 PM	1c033690-c955-498a-b463ad090a01	
3/5/2026, 3:19:54 PM	e982fe47-b616-406f...	
3/5/2026, 3:13:05 PM	37d72952-c775-4f51...	
3/5/2026, 3:13:04 PM	daa2ea2b-27ab-4f96...	
3/5/2026, 3:10:50 PM	397e1738-8f6f-495a...	
3/5/2026, 3:10:46 PM	654ff025-710b-4c7e...	
3/5/2026, 3:05:40 PM	dbeb9a40-8c92-4b3...	
3/5/2026, 3:05:35 PM	ae9aa4c0-1c97-479c...	
3/5/2026, 2:49:45 PM	649861f4-386c-4013...	
3/5/2026, 2:46:39 PM	db7899f1-c057-4a74...	
3/5/2026, 2:46:37 PM	a6821bd7-6ca1-43a...	
3/5/2026, 2:46:34 PM	1370cc1a-4b45-4a84...	
3/5/2026, 2:44:46 PM	40edcb3a-9255-46a...	
3/5/2026, 2:44:44 PM	c29e69ab-2160-4e0f...	
3/5/2026, 2:44:42 PM	40edcb3a-9255-46a...	
3/5/2026, 2:39:34 PM	a67dc500-6713-496...	
3/5/2026, 2:14:52 PM	35f51ed7-ed07-4089...	
3/5/2026, 2:11:13 PM	c6648d65-12a6-407...	
3/5/2026, 2:07:12 PM	68e7e1b4-e2b6-49b...	
3/5/2026, 2:00:55 PM	7277f5fd-1cc2-4b7f...	
3/5/2026, 1:58:25 PM	05bd7686-e87b-494...	
3/5/2026, 1:58:23 PM	59e6c4d9-cb44-4a6...	
3/5/2026, 1:58:17 PM	05bd7686-e87b-494...	
3/5/2026, 1:53:16 PM	4cad1c97-0764-439...	
3/5/2026, 1:46:57 PM	4d3b39da-7e94-407...	
3/5/2026, 1:46:29 PM	05bd7686-e87b-494...	
3/5/2026, 11:37:32 AM	c699e9df-4677-48ab...	
3/5/2026, 10:18:07 AM	c2c53dec-a49e-46ac...	
3/5/2026, 10:15:28 AM	9583d6b5-4848-494...	
3/5/2026, 10:15:02 AM	ef4c64af-2e85-4a4a...	

Activity Details: Sign-ins

Basic info	Location	Device info	Authentication Details	Conditional Access
Date		3/5/2026, 3:19:54 PM		
Request ID		1c033690-c955-498a-9d64-b463ad090a01		
Correlation ID		f744dc27-d530-4554-af68-a7a34e8c4f72		
Authentication requirement		Multifactor authentication		
Agent Type		Not Agentic		
Status		Success		
Continuous access evaluation		No		
Follow these steps:				
Troubleshoot Event				
Launch the Sign-in Diagnostic.				
1. Review the diagnosis and act on suggested fixes.				
User		Richard van der Els		
Username		richardvanderels@vanderelstest.onmicrosoft.com		
User ID		65f8befd-9634-4f8f-89fb-82cc0e1ea64f		
Sign-in identifier				
Session ID		002e91fa-37db-9706-230f-31afe7b59474		
App owner tenant ID		f8cdef31-a31e-4b4a-93e4-5f571e91255a		
Resource owner tenant ID				
User type		Member		
Cross tenant access type		None		
Application		OfficeHome		
Application ID		4765445b-32c6-49b0-83e6-1d93765276ca		
Resource		OfficeHome		
Resource ID		4765445b-32c6-49b0-83e6-1d93765276ca		
Resource tenant ID		2f7a5a5d-9e00-463e-a554-e8119b6f36a2		
Home tenant ID		2f7a5a5d-9e00-463e-a554-e8119b6f36a2		
Home tenant name				
Client app		Browser		
Client credential type		None		
Service principal ID				
Original transfer method		None		
Token Protection - Sign In Session		Unbound (statusCode: 1002)		
Service principal name				
Resource service principal ID				
Unique token identifier		kDYDHFxikmdZLrJTQkKAQ		
Token issuer type		Microsoft Entra ID		

Controleer de sign-in logs om te verifiëren dat gebruikers inloggen met een MFA en een certificaat

Vragen?

